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Cross-scale observations of expanding bifurcated current sheets in fast magnetic reconnection

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Jongsoo Yoo[1] ✉, Masaaki Yamada[1], Tongnyeol Rhee[1,2], Hantao Ji[1,3], Ari Le[4], Tai Phan[5], Sung Hyun Son[3] & Adam Robbins[3]

Magnetic reconnection is a universal process that rapidly releases magnetic energy in plasmas, driving explosive phenomena in both space and laboratory environments. Yet a complete understanding of how reconnection proceeds in large-scale plasmas remains elusive. Here we show that expanding bifurcated current sheets (EBCS) are a recurring structure of quasi-steady reconnection. EBCS are double-peaked current layers that expand along the boundaries of the reconnection outflow, accompanied by localized electric fields and strong ion acceleration. We identify this structure consistently in laboratory experiments, spacecraft observations near Earth and the Sun, and large-scale kinetic simulations. Our results indicate that EBCS arise from two-fluid plasma dynamics and occur during quasi-steady antiparallel reconnection, particularly for plasmas with moderate plasma beta ($0.1 < \beta \lesssim 1$, where β is the ratio of plasma thermal pressure to magnetic pressure). These findings link laboratory and space observations to the classical Petschek model and show how two-fluid dynamics can produce open reconnection outflows across widely different system sizes.

The vast majority of visible matter in the Universe exists in a plasma state, constantly interacting with embedded magnetic fields. A fundamental process in such magnetized environments is magnetic reconnection, which converts magnetic energy into plasma kinetic energy and drives phenomena such as solar eruptions and geomagnetic storms. Magnetic reconnection is observed across systems ranging from astrophysical environments such as solar flares to Earth's magnetosphere and laboratory fusion devices, where it plays a central role in rearranging magnetic topology and transporting plasma. In fusion devices, reconnection also influences the stability and energy dynamics of magnetically confined plasmas[1–3].

Early magnetic reconnection research focused on the so-called fast reconnection problem, aiming to explain the rapid reconnection rates observed in phenomena such as solar flares and sawtooth relaxations in fusion devices—rates that exceed predictions from classical resistive models such as Sweet–Parker reconnection[4,5]. Petschek addressed this problem by proposing an open exhaust geometry with standing shocks that enable rapid, Alfvénic plasma outflows[6]. While Petschek-like open exhausts have often been inferred from space and laboratory observations under low-resistivity conditions [e.g., refs. 7–9], direct evidence for the predicted switch-off slow shocks bounding collisionless reconnection exhausts has remained inconclusive[10].

Expanding bifurcated current sheets (EBCS) have been reported in heliospheric current sheets[7,11–15] and Earth's magnetotail[16,17]. These structures are termed expanding because the distance between the double-peaked current layers increases downstream as they follow the diverging separatrices. Here, EBCS are defined as double-peaked current layers generated by two-fluid effects that extend outward along magnetic separatrices. In symmetric collisionless reconnection, these

[1]Princeton Plasma Physics Laboratory, Princeton, NJ, USA. [2]Korea Institute of Fusion Energy, Daejeon, Republic of Korea. [3]Department of Astrophysical Sciences, Princeton University, Princeton, NJ, USA. [4]Los Alamos National Laboratory, Los Alamos, NM, USA. [5]Space Sciences Laboratory, University of California, Berkeley, CA, USA. ✉e-mail: jyoo@pppl.gov

bifurcated sheets initially form near the X-line as a consequence of kinetic-scale electron–ion decoupling. Their macroscopic spatial extent, however, depends on the global plasma conditions. In bursty, low-β environments, where β is the ratio of plasma thermal pressure to the magnetic pressure, the boundary can be disrupted by secondary instabilities and plasmoid formation. Under quasi-steady conditions, these kinetic boundaries can instead persist far downstream and form an open exhaust geometry resembling that envisioned by Petschek[6]. Throughout this manuscript, the term EBCS is used specifically for these macroscopically extended boundaries that persist far downstream, with an extent comparable to the system size.

EBCS are associated with sharp magnetic-field variations, localized electric fields, and enhanced ion outflows. These signatures resemble features of the open exhaust geometry envisioned by Petschek[18,19], but the mechanisms governing EBCS formation and their relation to reconnection dynamics remain incompletely understood[3].

The term quasi-steady refers to reconnection in which the global reconnection rate[20], when averaged over several Alfvén times, remains approximately constant over an extended period. High-resolution observations and simulations show that electron-scale dynamics near the central X-line can remain bursty or turbulent even when the macroscopic exhaust persists over longer temporal and spatial scales. For example, a recent space observation showed that a Petschek-type exhaust bounded by sharp discontinuities can persist far downstream despite turbulent dynamics in the central diffusion region[21]. We therefore distinguish the global quasi-steady evolution of the exhaust from local, time-dependent dynamics near the X-line.

In this work, we provide experimental, observational, and numerical evidence that clarifies the origin of EBCS and their connection to reconnection physics. We show that EBCS emerge during quasi-steady reconnection, particularly in plasmas with moderate β (the ratio of plasma thermal pressure to magnetic pressure), and are generated by two-fluid effects, in which differential electron and ion dynamics establish electrostatic potential wells and strong in-plane electric fields along separatrices. Large-scale kinetic simulations, together with laboratory and spacecraft measurements, show that EBCS occur in quasi-steady reconnection exhausts across the system sizes examined here and connect Petschek-like open exhaust geometries with two-fluid plasma dynamics.

Results

Two-fluid origin and structure of EBCS

As shown in Fig. 1, the two-fluid model of magnetic reconnection highlights the different behaviors of electrons (dashed red lines) and ions (dashed blue lines) within the collisionless reconnection layer. As oppositely directed magnetic field lines converge at the X-line, both species are drawn into the reconnection region, but a critical difference in their magnetization arises.

Ions demagnetize first, forming a broad ion diffusion region (IDR, gray box), where their motion is no longer tied to the magnetic field. Electrons, in contrast, remain magnetized until reaching the much narrower electron diffusion region (EDR, yellow box), where they too become demagnetized. This difference allows magnetized electrons to penetrate more deeply than ions, producing charge separation and a strong localized electrostatic potential well near the midplane[3,22–25].

This potential structure creates a deep potential well that spans the entire macroscopic exhaust region. However, the steep potential drops (and therefore the strongest in-plane electric fields, primarily oriented along the inflow direction, E_x) are highly localized to the narrow separatrix regions bounding the exhaust. Magnetized electrons respond to this localized electric field by undergoing a strong $\mathbf{E} \times \mathbf{B}$ drift specifically within this boundary layer. This localized drift, coupled with the reconnecting magnetic field component (B_z), is the mechanism that generates a strong out-of-plane electron current density specifically within the separatrix region, creating EBCS (orange).

It is important to clearly distinguish the narrow separatrix region from the broader exhaust region it encloses. The formation of the EBCS is localized to the separatrix region, where the sharp potential drop causes a sudden change in the magnetic field direction and drives a sharp acceleration of ions across the boundary. Once accelerated across this narrow EBCS layer, the resulting bulk ion outflow and density enhancements subsequently fill the entire macroscopic exhaust region. To illustrate these phenomena in detail, we analyze a two-dimensional (2D) particle-in-cell (PIC) simulation of antiparallel reconnection using the high-performance VPIC (Vector Particle-In-Cell) code[26]. The simulation employs a physical ion-electron mass ratio of 1836, an upstream β of 0.16, and a domain size of $20d_i \times 20d_i$, where d_i is the ion skin depth. The normalized reconnection rate for this simulation is approximately 0.1. Further details of the Harris sheet setup are provided in the Methods section and in ref. 27.

Figure 2 presents simulation results that collectively demonstrate the formation of EBCS and associated features in collisionless antiparallel magnetic reconnection. Fig. 2a shows the out-of-plane current density J_y, which exhibits a clear bifurcated profile near the separatrix. Figure 2b shows the electrostatic potential Φ, which reveals a deep well centered near the reconnection midplane. The potential well is directly related to characteristics of EBCS. Simultaneously, this structure accelerates ions across the separatrix, as shown in Fig. 2c, which plots the ion outflow velocity V_{iz}. A sharp spatial gradient in the ion outflow velocity, indicating strong, localized acceleration, is observed directly across the separatrix layer. Once accelerated across this EBCS boundary, the high-velocity ions fill the broader exhaust region. This

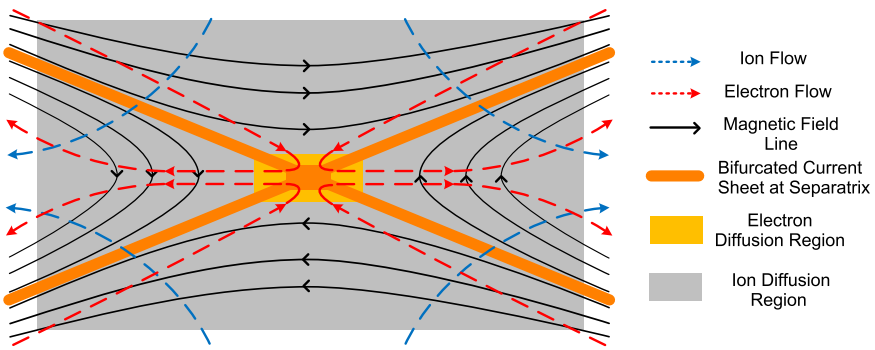


Fig. 1 | Schematic of a prototypical magnetic reconnection layer. Due to their different kinetic scales, ions decouple from electrons in the ion diffusion region (gray), while electrons remain frozen-in until reaching the electron diffusion region (yellow). This decoupling gives rise to a strong in-plane electric field near the separatrix. Ions are accelerated into the exhaust region by this electric field, while electrons form a kinetic-scale bifurcated current sheet near the separatrix via the $\mathbf{E} \times \mathbf{B}$ drift[3].

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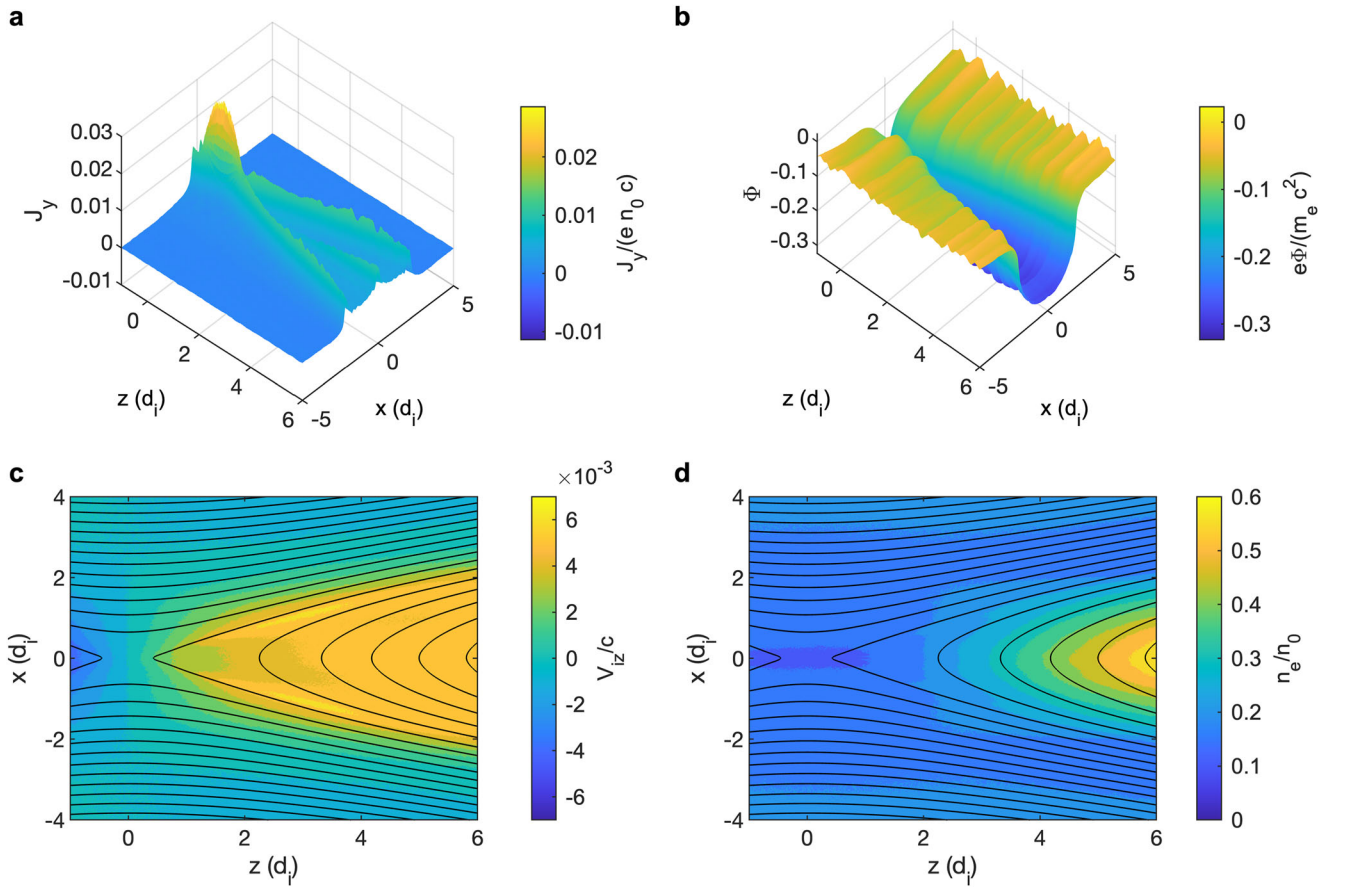


Fig. 2 | Simulation results illustrating key features related to expanding bifurcated current sheets (EBCS). All quantities are normalized, and spatial coordinates are given in units of the ion inertial length (d_i). **a** Out-of-plane current density (J_y), normalized to en_0c, where e is the elementary charge, n_0 is the reference density, and c is the speed of light, showing the bifurcated structure near the separatrix. **b** Electrostatic potential (Φ), normalized to m_ec^2/e, where m_e is the electron mass, revealing the formation of a deep potential well. **c** Ion outflow velocity (V_{iz}), normalized to c and overlaid with magnetic field lines, showing a sharp enhancement across the separatrix consistent with Petschek-type reconnection. **d** Electron density (n_e), normalized to n_0, showing density enhancement in the exhaust region.

localized cross-boundary acceleration is a kinetic characteristic of a Petschek-type exhaust.

Finally, Fig. 2d displays the plasma density profile, which shows a pronounced enhancement in the exhaust region. This density increase is a natural consequence of ion acceleration into the exhaust; ion acceleration by the in-plane electric field creates a converging ion flow pattern in the exhaust region. Together, these panels demonstrate that EBCS are not an isolated feature but rather a key organizing structure that governs the dynamics of magnetic reconnection and energy conversion.

Laboratory observation of EBCS

To verify the existence of EBCS and their associated structures during magnetic reconnection, we first turn to measurements from the Magnetic Reconnection Experiment (MRX)[28]. MRX is a laboratory platform dedicated to studying magnetic reconnection, consisting of a cylindrical vacuum vessel 1.5 m in diameter and 2 m in length, in which reconnection is driven using two sets of internal coils. A two-dimensional (2D) magnetic probe array, comprising more than 100 spatial points, is employed to map the full 2D magnetic field vector. To resolve fine-scale features that the array cannot capture, a one-dimensional (1D) magnetic probe with significantly higher spatial resolution is deployed in targeted regions near the separatrix.

Complementary plasma diagnostics provide additional information on magnetic reconnection. Electron temperature and density are measured using a triple Langmuir probe[29], while ion outflow velocities are obtained from a Mach probe[30]. A floating potential probe array yields high-resolution 1D profiles of the electrostatic potential. For analysis, we adopt a local xyz coordinate system consistent with Fig. 1, where x corresponds to the radial direction in MRX and z to the axial (outflow) direction. All lengths are normalized to the ion skin depth (d_i = 2.5 cm), defined using the upstream density on the high-density side. MRX typically operates in a regime where collisionless dynamics dominate[31,32], and the plasma conditions across the current sheet are asymmetric: the high-density side exhibits β of approximately 1.1, whereas the low-density side has β of approximately 0.2[30]. The normalized reconnection rate for this quasi-steady MRX regime is fast, with R approximately 0.15.

Figure 3 presents simultaneous laboratory measurements of the structures associated with EBCS. Figure 3a shows the 2D profile of the out-of-plane current density J_y, overlaid with magnetic field lines. The 2D current profile shown in Fig. 3a is obtained from direct measurements of the full magnetic field vector. Using the 2D magnetic probe array, the magnetic field components are simultaneously measured across a 2D spatial grid. The 2D out-of-plane current density (J_y) is then directly calculated from the measured in-plane magnetic fields (B_x and B_z) using Ampère's law, $J_y = (\partial B_x/\partial z - \partial B_z \partial x)/\mu_0$, where μ_0 is the vacuum permeability. While hints of current sheet bifurcation are apparent near the separatrix, the array resolution is insufficient to resolve narrow current layers. This limitation is overcome using the high-resolution 1D magnetic probe, positioned at $z = 6d_i$ to capture the detailed structure of EBCS.

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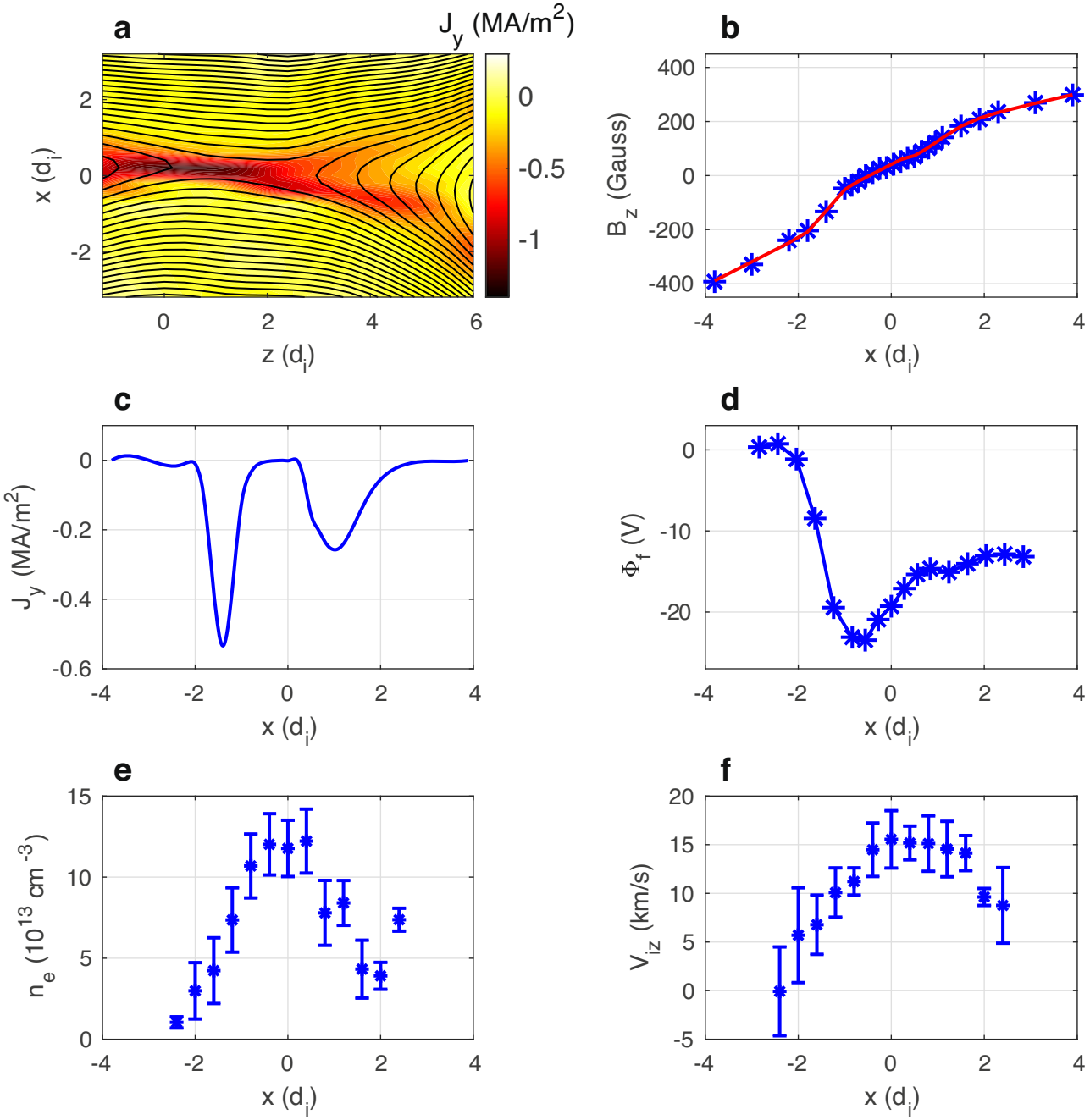


Fig. 3 | Experimental measurements from the Magnetic Reconnection Experiment (MRX) showing key features associated with EBCS. All spatial coordinates are normalized to the ion skin depth d_i = 2.5 cm, calculated from the upstream density on the high-density side. **a** Two-dimensional profile of the out-of-plane current density J_y, demonstrating EBCS. The black lines represent magnetic fields. **b** One-dimensional profile of the reconnecting magnetic field B_z at $z = 6d_i$. The blue asterisks denote the experimental measurements and the red curve denotes the double-Harris-sheet fit. **c** One-dimensional profile of J_y, derived from the gradient of the measured B_z, confirming the presence of two distinct current peaks. **d** Floating potential (Φ_f) profile, indicating an electrostatic potential well at $z = 6d_i$. **e** Electron density profile showing enhancement in the exhaust region. **f** Ion outflow velocity (V_{iz}), highlighting acceleration across the separatrix. Error bars represent the standard deviation of the data.

Figure 3 b shows the reconnecting magnetic field component B_z at $z = 6d_i$, measured by the high-resolution probe. The blue asterisks denote experimental data, while the red curve corresponds to fitting to a double Harris-sheet (a sum of two tanh functions)[33]. The left layer is characterized by the profile $49\tanh[(x+1.4d_i)/0.29d_i]$ Gauss, while the right layer is fitted to $57\tanh[(x-1.0d_i)/0.70d_i]$ Gauss. The steep gradients at approximately $x = -1.4d_i$ and $x = 1.0d_i$, together with the flattened profile in the central region, confirm the presence of EBCS. To obtain the plasma current density J_y, the measured B_z is fitted with a profile consisting of a double-Harris sheet and a linear background term. This linear term represents the vacuum magnetic field from the external coils, which carries a gradient but no local plasma current[34]. By taking the derivative of only the tanh components of the fit, we obtain the plasma current density shown in Fig. 3c. The profile has two distinct peaks around $x = -1.4d_i$ and $1.0d_i$. The asymmetry between the peaks reflects the upstream density gradient; as shown in Fig. 3e, the upstream density on the high-density side ($x > 0$) is approximately five times greater than on the low-density side ($x < 0$).

Additional diagnostics further corroborate the EBCS structure. Figure 3d shows the floating potential (Φ_f) profile. Because variations in electron temperature across the separatrix are limited to about 1 eV, Φ_f can be used as a direct proxy for the plasma potential. The resulting profile reveals a distinct, asymmetric potential well whose minimum is shifted toward the low-density side, at approximately $x = -0.8d_i$. The potential exhibits a sharp drop near $x = -1.5d_i$ on the low-density side, whereas the change on the high-density side is much more gradual. Furthermore, there is a noticeable baseline difference of about 13 V between the two upstream regions; this offset is determined by the global density asymmetry, which drives a macroscopic ambipolar electric field across the entire current sheet layer[31]. The inferred in-plane electric field (about 900 V/m) is sufficient to drive $\mathbf{E} \times \mathbf{B}$ drifts comparable to the measured electron velocities (approximately 60 km/s near $x = -1.4d_i$).

The electron density profile in Fig. 3e, obtained by scanning a Langmuir probe across 250 discharges under identical operating conditions, reveals a pronounced asymmetry between the upstream regions. A sharp density increase is observed across the separatrix, accompanied by a small but distinct dip near $x = 2d_i$. This dip reflects a local reduction in density where ion acceleration begins (see Fig. 2d). The resulting divergent ion flow pattern reduces the plasma density, producing the observed dip. On the low-density side, however, no such feature is present, since the density asymmetry prevents the ion flow pattern from diverging there[31].

Lastly, Fig. 3f presents the one-dimensional profile of ion outflow velocity (V_{iz}). The ion flow increases as ions are accelerated by the in-plane electric field near the separatrix, reaching a peak near the mid-plane, which is consistent with the expected reconnection outflow profile.

Spacecraft observation of EBCS in Earth's Magnetotail

Having established the presence of EBCS and associated signatures in the controlled laboratory setting of MRX, we next turn to observations in a natural plasma environment to test their universality. Figure 4 presents in situ measurements from the Magnetospheric Multiscale (MMS) mission during a magnetotail reconnection event on June 19, 2017. Unlike MRX, this event exhibits relatively symmetric upstream conditions, with β of approximately 0.6 on both sides of the current sheet. Nevertheless, MMS reveals similar signatures near the separatrix, including EBCS-like magnetic field structures, sharp density enhancements, accelerated ion outflows, and strong normal electric fields. The black dashed vertical lines in Fig. 4 mark the estimated separatrix crossings, showing MMS traversing the reconnection exhaust between them. Multi-spacecraft timing analysis[35,36] indicates a current sheet normal velocity of about 40 km/s, corresponding to an exhaust width of about $20d_i$ (d_i is approximately 700 km) and implying an X-line distance of about $100d_i$ for a typical aspect ratio of the reconnection layer of 5.

Figure 4a shows all three magnetic field components during the crossing. Distinct variations in B_z are seen near 09:46:05 and 09:50:45, while a plateau-like feature is observed near 09:48:00 in the middle of the exhaust region. These signatures, analogous to those identified in MRX, provide compelling evidence for EBCS in Earth's magnetotail.

Figure 4b displays the electron density (n_e), which shows no significant asymmetry between the upstream regions, consistent with the nearly symmetric plasma conditions of this event. Notably, localized density dips are observed at both separatrix crossings, a feature that mirrors the MRX results (Fig. 3e) as well as other MMS observations[37]. These density dips are considered reliable signatures of the separatrix, particularly in cases where the spacecraft passes close to the X-line[37].

Figure 4c presents the ion velocity components (V_{ix}, V_{iy}, and V_{iz}). Both density and ion outflow velocity increase in the exhaust region, consistent with MRX observations. In this event, the value of V_{iz} is negligible in the upstream region and reaches approximately 300 km/s in the exhaust region. This acceleration reflects efficient ion energization by the in-plane electric field, and the overall pattern closely matches both laboratory and simulation results.

The normal electric field component E_x, shown in Fig. 4d, provides additional evidence for the presence of an electrostatic potential well. The red trace shows the low-pass-filtered signal to emphasize large-scale features. A pronounced negative excursion in E_x is observed near the first separatrix crossing (gray box), while the second crossing displays two smaller but distinct positive peaks (gray boxes). In both cases, the electric field points inward toward the exhaust, consistent with the presence of an electrostatic potential well.

Figure 4e displays the out-of-plane electron current density (blue line, J_{ey}), which is low-pass filtered to remove high-frequency noise. To verify the physical origin of this current, we also plot the out-of-plane current density expected purely from the local $\mathbf{E} \times \mathbf{B}$ drift (red line, $-en_eV_{E\times B,y}$), where e is the positive elementary charge and n_e is the electron density. The results show that J_{ey} is not concentrated at the exhaust center, but instead enhanced near the separatrices, consistent with the EBCS structure. Moreover, intervals of strong E_x (gray boxes in panels d and e) coincide with these current enhancements. The agreement between the measured electron current and the calculated drift current in these bounded regions suggests that the out-of-plane electron current is indeed primarily driven by the $\mathbf{E} \times \mathbf{B}$ drift within the electrostatic potential well.

Notably, the middle J_{ey} peak does not produce a step-like change in B_z. This likely indicates a transition from a quasi-1D current sheet into a complex 2D or 3D structure driven by instabilities like lower hybrid drift instability (LHDI). In such multidimensional regions, the 1D relationship between magnetic gradients and out-of-plane current is locally violated, allowing for significant current density without the macroscopic B_z signatures observed at the primary separatrix crossings. Additionally, the current sheet undergoes bifurcation near the second separatrix, producing a double-peak structure that corresponds to the double E_x feature observed at the same location. Together, these observations indicate that complex, multi-layered current structures can develop near the separatrix. As the MMS event features the lowest electron beta of our three examples, with β_e (the ratio of electron thermal pressure to magnetic pressure) approximately 0.17, the current sheet is highly susceptible to secondary instabilities, such as LHDI. This results in the observed fluctuations in E_x and the further fragmentation of the EBCS into the minor secondary peaks visible in Fig. 4e. This demonstrates that while the EBCS is the primary organizing boundary, it can exhibit multi-layer sub-structures in low-β environments.

The ion temperature profile (T_i), shown in Fig. 4f, rises from approximately 3 keV in the upstream region to about 5 keV within the exhaust. This notable increase is attributed to counter-streaming ion beams produced as ions are accelerated by the normal electric field near the separatrix[15,24,38]. Finally, Fig. 4g shows the z (outflow) component of the ion flow energy flux, decomposed into the bulk kinetic contribution ($K_{iz} = m_in_iV_i^2V_{iz}/2$; m_i is the ion mass, n_i is the ion number density, and V_i is the ion bulk speed) and the modified ion enthalpy flux ($H_{iz} = 5n_i(T_i - T_{i,up})V_{iz}/2$; $T_{i,up}$ is the upstream ion temperature). Both quantities are evaluated in the frame where V_{iz} is approximately zero in the upstream region. The subtraction of the upstream ion temperature ensures that H_{iz} represents only the additional heating due to reconnection dynamics, enabling a direct comparison with K_{iz}. In this event, the enthalpy flux dominates over the bulk kinetic flux, even after subtracting the upstream temperature of 2.8 keV. While instrumental energy-range limitations of the FPI (Fast Plasma Investigation) can introduce some uncertainty into the absolute ion temperature moments, this dominance of enthalpy flux is physically required by the overall energy budget. The potential well depth for this event is estimated to exceed 5 kV (Table 1), yet the bulk ion outflow of <400 km/s accounts for much less than 1 keV of kinetic energy per proton.

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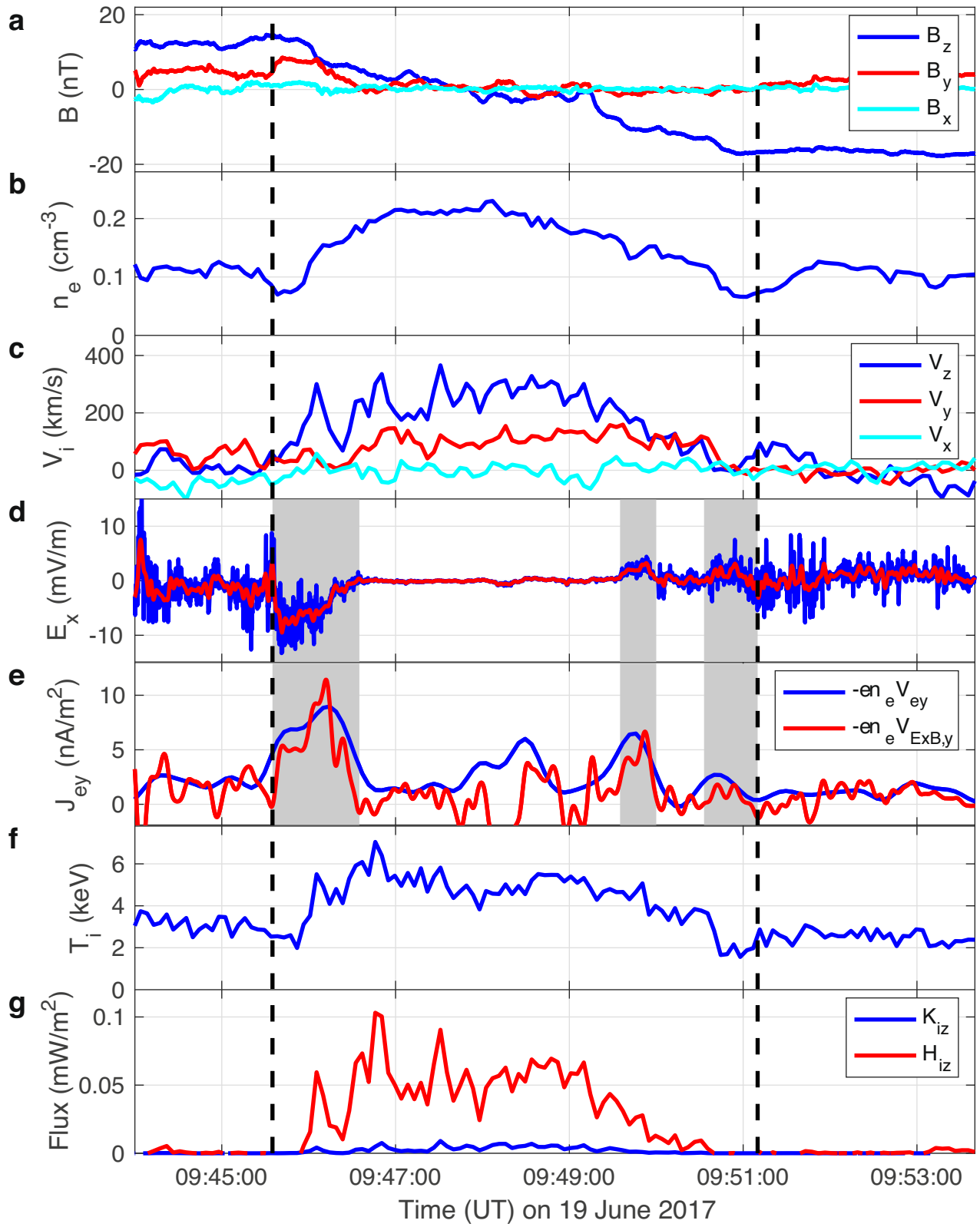


Fig. 4 | Magnetic and plasma measurements from MMS during a reconnection event in Earth's magnetotail on 19 June 2017. Two dashed vertical lines mark the locations of the separatrix. **a** Magnetic field (**B**) in the local boundary-normal coordinate system (determined via Minimum Variance Analysis; see "Methods" for the precise transformation from GSE coordinates). The B_z profile shows step-like changes near the separatrix while it is flat in the middle of the exhaust region, which is consistent with EBCS. **b** Electron number density (n_e), showing increase across EBCS. **c** Ion flow velocity ($\mathbf{V}_i$), showing an increase in the V_z component across EBCS. **d** Electric field component along the x direction (E_x), showing strong in-plane electric fields near the separatrix. The red line represents the data processed with a low-pass filter. The gray boxes indicate the time with the notable E_x in the exhaust region. **e** Out-of-plane electron current density (blue, $-en_eV_{ey}$) compared with the $\mathbf{E} \times \mathbf{B}$ drift current density (red, $-en_eV_{E\times B,y}$). Both are low-pass filtered. The close agreement between the two traces during the times of sizable E_x (gray boxes) confirms that the EBCS is primarily driven by the $\mathbf{E} \times \mathbf{B}$ drift. **f** Ion temperature (T_i), which exhibits a clear increase in the exhaust region. **g** The z component of the ion flow energy flux (K_{iz}) and enthalpy flux (H_{iz}). A baseline (upstream) temperature of 3 keV is subtracted from H_{iz} for clarity. The outgoing ion energy is dominated by the enthalpy flux.

Consequently, the vast majority of the energy gained by ions traversing the potential drop must be converted into thermal heating and plasma compression. This indicates that the majority of ion energization takes the form of thermal heating rather than bulk acceleration. Such behavior is consistent with earlier observations in Earth's magnetotail[39] as well as laboratory results[40].

Spacecraft observation of EBCS in the heliospheric current sheet

To extend the cross-environment comparison into a broader heliophysical context, we examine in situ measurements from the Parker Solar Probe (PSP). Despite the enormous difference in system size (about $10d_i$ in MRX versus about $10^4 d_i$ in the heliospheric current sheet), the key features observed near the separatrix such as EBCS, sudden density enhancements, and large ion outflows are again evident. Historically, such macroscopic boundaries in the solar wind have often been evaluated using one-dimensional (1D) ideal magnetohydrodynamic (MHD) Rankine–Hugoniot conditions to classify them as slow mode shocks[6,21]. However, because actual reconnection exhausts are inherently two dimensional with expanding separatrices, strict 1D planar conservation laws are rarely perfectly satisfied. In this study, rather than treating these boundaries as 1D macroscopic fluid discontinuities, we focus on resolving the underlying kinetic scale physics: the EBCS structures that fundamentally mediate the transition layer. Figure 5 shows data from a reconnection event on April 29, 2021 (PSP Encounter 08)[15], with black dashed vertical lines marking the estimated separatrix crossings. During this interval, PSP traversed the reconnection exhaust, allowing a direct comparison with the MRX and MMS observations. The plasma β is asymmetric across the current sheet, with a value of approximately 0.6 on the low-density side and approximately 1.7 on the high-density side.

Figure 5a shows the three components of the magnetic field. Similar to the MRX observations (Fig. 3b), sharp decreases in the reconnecting field component (B_z) are observed near the separatrix. These abrupt changes mark the presence of EBCS, whose thickness (about $20d_i$; d_i is approximately 4 km) is much narrower compared to the overall exhaust width of about $4000d_i$[15].

Table 1 | Comparison of the potential well magnitudes estimated from Eq. (4) with the measured values

	Low-density (Estimated)	Low-density (Measured)	High-density (Estimated)	High-density (Measured)
MRX	12 V	11 V	6 V	5 V
MMS[a]	6 kV	5 kV	5 kV	7 kV
PSP[b]	120 V	N/A	60 V	N/A

For both MRX and MMS cases, the estimates from the simple formula are in good agreement with the measurements.
[a]The density profile of the MMS event is symmetric; the low-density side corresponds to the separatrix encountered first.
[b]The uncertainty in PSP electric field measurements is too large to allow a reliable determination of the potential well magnitude.

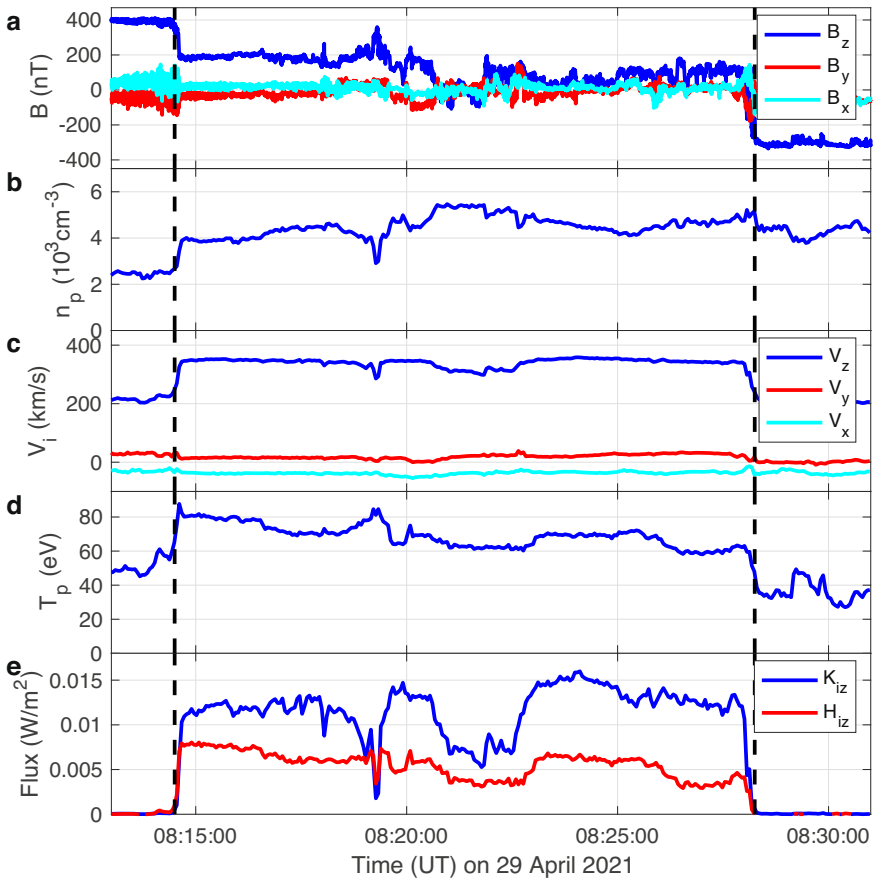


Fig. 5 | Magnetic and plasma measurements from Parker Solar Probe during a heliospheric current sheet crossing. Two dashed vertical lines mark the locations of the separatrix. **a** Magnetic field (**B**) in the local magnetic coordinate system. A sharp reduction in the B_z component near the separatrix indicates the presence of EBCS. **b** Proton number density (n_p), which increases abruptly across EBCS. The upstream density asymmetry is moderate, with a density ratio of approximately 2. **c** Ion flow velocity ($\mathbf{V}_i$), showing a sudden increase in the V_z component across EBCS. **d** Proton temperature (T_p), which exhibits a step-like increase at the EBCS crossing. **e** The z component of the ion flow energy flux (K_{iz}) and enthalpy flux (H_{iz}). Here, K_{iz} is calculated in the upstream ion rest frame, and a baseline temperature of 50 eV is subtracted from H_{iz} for clarity.

Figure 5b displays the proton density (n_p, used here rather than electron density, as solar wind electron measurements are often subject to photoelectron contamination), which shows a sudden increase at 08:14:30 when PSP crossed the separatrix on the low-density side, and a sharp decrease at 08:28:15 when the spacecraft exited into the high-density upstream. This modest density asymmetry (a factor of about 2) mirrors the laboratory findings (Fig. 3e) and emphasizes that even in the heliosphere, reconnection exhausts can exhibit the same signatures of asymmetric reconnection seen in controlled experiments.

The ion velocity components are shown in Fig. 5c. The ion outflow along the z direction increases rapidly inside the exhaust, with its onset coinciding with the EBCS locations. This behavior is consistent with MRX results, where strong in-plane electric fields near the separatrix were inferred from floating potential measurements (Fig. 3d) and directly linked to ion acceleration.

Figure 5d presents the proton temperature (T_p), which rises from about 50 eV in the upstream regions to about 70 eV in the exhaust. The sharp increase occurs at the EBCS crossings, implying that part of the energy gained from the in-plane electric field is converted directly into ion heating. This result aligns well with the MMS observations, where ion heating is also observed across the separatrix, though with different relative contributions of bulk versus thermal energy.

The energy partition is quantified in Fig. 5e, which shows the z component of the ion energy fluxes: the bulk flow contribution (K_{iz}) and the modified enthalpy flux (H_{iz}). Unlike the MMS case, where enthalpy flux dominates, here the ion flow energy flux exceeds H_{iz}. This indicates that in the heliospheric current sheet, a greater fraction of magnetic energy is converted into organized ion outflow (jets) rather than thermal energy. Such a distinction is notable because, in many reconnection events, heating is the dominant channel of ion energization[39,40].

This distinction is likely related to the electron beta (β_e), which serves as a control parameter for the lower hybrid drift instability (LHDI) near the separatrix[41–43]. For the MMS case, the upstream β_e (0.17) is significantly lower than the PSP case (≥0.6), creating conditions favorable for LHDI development[41,44,45]. The fluctuations in E_x shown in Fig. 4d near the two separatrices are associated with this instability; wavelet analysis of E_x reveals broadband fluctuation energy concentrated predominantly just below the local lower hybrid frequency of approximately 8 Hz, which is a characteristic signature of LHDI[44,46]. The presence of LHDI makes it more difficult to maintain the clear boundary structure as observed in the PSP case, resulting in enhanced ion heating through phase mixing[41].

Electrostatic potential wells and energy conversion in EBCS

The consistent appearance of EBCS, potential wells, and high-density exhausts across different plasma environments and over a large range of the system size highlights the universality of magnetic reconnection dynamics. These structures play a crucial role in energy conversion. In particular, the structure of the separatrix region and its associated electric fields appear central to directing magnetic energy into particle acceleration and heating. We now turn to how these features are related to energy conversion during reconnection.

The depth of the potential well associated with the bifurcated current sheet can be estimated from the electron equation of motion in the x direction[25,47]:

$$E_x \approx V_{ey}B_z - V_{ez}B_y - \frac{1}{en}\frac{\partial p_e}{\partial x}, \tag{1}$$

where E_x is the normal electric-field component, V_{ey} and V_{ez} are the electron velocity components, B_y and B_z are the magnetic-field components, e is the elementary charge, n_e is the electron number density, p_e is the scalar electron pressure, and x is the current-sheet normal coordinate. The pressure gradient term is typically less than 10%, indicating that the in-plane electric field is largely governed by ideal electron fluid dynamics. Physically, the $\mathbf{V}_e \times \mathbf{B}$ term represents the change in the magnetic pressure across the boundary, while the pressure gradient term represents that of the electron pressure. Because the energy from the deep potential well predominantly drives strong ion heating and acceleration[38], the electron temperature does not increase significantly[25]. Consequently, the change in magnetic pressure largely exceeds the change in electron pressure, making the electron diamagnetic contribution small. Since $V_e \gg V_i$ in EBCS, we evaluate this in the ion frame, i.e., $V_i = 0$. From Ampère's law:

$$V_{ey} \approx -\frac{J_y}{en} \approx \frac{1}{en\mu_0}\frac{\partial B_z}{\partial x}, \tag{2}$$

assuming $\partial B_x/\partial z \ll \partial B_z/\partial x$ within EBCS. Here, J_y is the out-of-plane current density. Also, since $\partial/\partial y = 0$, we have:

$$V_{ez} = -\frac{1}{en\mu_0}\frac{\partial B_y}{\partial x}. \tag{3}$$

Integrating Eq. (1) across EBCS yields the depth of the potential well (Φ_0):

$$e\Phi_0 \approx \frac{\Delta B_z^2 + \Delta B_y^2}{2\mu_0 n_{av}}. \tag{4}$$

Here, ΔB_y and ΔB_z are the changes in the corresponding magnetic-field components across the EBCS, and n_{av} is the density averaged over the region with significant current density. Treating the density as an average constant across this specific layer is a practical simplification. Integrating the exact density profile yields values that differ from this simple formula by only about 10%. Given the typical uncertainties in measuring electric fields and densities, this formula provides a highly reliable, straightforward proxy for the potential drop and clearly highlights the physical scaling.

Table 1 summarizes the performance of Eq. (4) in comparison with measured values. For both MRX and MMS cases, the estimates from the simplified expression agree well with the measurements, indicating that the model provides a reliable approximation of the potential well magnitude. In MRX, the potential well magnitude is obtained from the difference in floating potential across the separatrix. For MMS, the time-series E_x profile is converted into a spatial potential using an estimated normal velocity of 40 km/s. The comparison is not available for PSP data due to the large uncertainties in the electric field measurements.

Equation (4) also explains why ions typically gain more energy than electrons in magnetic reconnection. Usually ΔB_y^2 is negligible compared to ΔB_z^2, and n_{av} is close to the upstream density (n_{up}). Then, Eq.(4) reduces to $e\Phi_0 = m_i V_A^2/2$, where V_A is the upstream Alfvén speed. Since $m_i V_A^2$ represents the available magnetic energy per ion-electron pair[48], this suggests that ions receive approximately half the incoming energy from the potential well. When additional energization from the out-of-plane reconnection electric field across the ion diffusion region is included, which is often comparable to the energy gain from the in-plane field, ions ultimately receive more than half of the available magnetic energy.

Equation (4) also highlights the relationship between the potential well magnitude and plasma beta. When ΔB_y^2 is negligible compared with ΔB_z^2 and n_{av} is approximated by the upstream density n_{up}, normalizing $e\Phi_0$ by the sum of the upstream electron and ion

temperatures yields

$$\frac{e\Phi_0}{T_e + T_i} \approx \frac{1}{\beta'} \quad (5)$$

where T_e and T_i are the upstream electron and ion temperatures, respectively, and β is the total upstream plasma beta. This relation indicates that stronger ion heating and acceleration are expected in low-β reconnection. Furthermore, this also reveals an upper limit for EBCS formation: in high-β regimes ($\beta \gg 1$), the potential well becomes too shallow to have distinct EBCS formation. Conversely, in low-β regimes, while the well is deep, the current sheet is highly susceptible to fragmentation via secondary instabilities (like LHDI and plasmoids). Thus, coherent EBCS are optimally formed and maintained in an intermediate, moderate-β regime ($0.1 < \beta \lesssim 1$), consistent with the conditions of our three analyzed events.

While Eqn. (5) establishes the macroscopic magnitude of the potential well, the cross-field spatial scale of the potential drop evolves dynamically. The charge separation generating this potential originates near the X-line, where the initial thickness is strictly constrained to the narrow electron kinetic scale. Under quasi-steady conditions, highly mobile electrons map this potential structure far downstream along the separatrices. However, as the exhaust extends macroscopically, the boundary naturally broadens from sub-d_i scales in MRX, to several d_i in MMS, to roughly $20d_i$ in the PSP event. Because this potential profile is fundamentally governed by electron dynamics, this spatial broadening is driven by electron finite gyro-radius effects; as electrons deliver the potential information further downstream, their cross-field motion gradually weakens the sharpness of the spatial gradient across the magnetic field. This steep, though evolving, spatial gradient, combined with the large potential magnitude, results in the localized in-plane electric field that drives the $\mathbf{E} \times \mathbf{B}$ current of the EBCS. Conversely, if the current sheet is unstable to secondary instabilities such as plasmoid formation[49] or the lower hybrid drift instability (LHDI)[50], the coherent transmission of this potential information along the magnetic field is disrupted, thereby fundamentally limiting the downstream extension of EBCS.

Large-scale kinetic simulations of EBCS

To cross-validate our laboratory and space-based findings, we performed additional VPIC simulations with a larger domain size ($200d_i \times 40d_i$) to investigate whether EBCS structures can persist over extended distances, as observed in space. The simulations use a reduced mass ratio ($m_i/m_e = 100$) and an upstream ion-to-electron temperature ratio of unity, consistent with conditions relevant to both MRX and PSP. To mitigate transient effects from initial condition, the initial Harris current sheet is given a broad width of $5d_i$. To reduce computational cost, the plasma-to-cyclotron frequency ratio is set to 2, which is smaller than the physical value ($\lesssim$100). Open-boundary conditions are implemented along the z direction[51], allowing plasma and electromagnetic fluxes to exit the simulation domain and thereby achieving quasi-steady reconnection.

Figure 6 summarizes the results of this large-domain simulation. The upstream plasma beta is 1.2, representative of PSP observations. Figure 6a shows the development of EBCS near the separatrix. While the macroscopic exhaust naturally widens and evolves as it propagates downstream along the z-direction, the kinetic EBCS boundary structure persists coherently, extending to the boundary of the simulation domain. This demonstrates that EBCS structure can persist far downstream from the X-line, consistent with the spatial extent observed by PSP and MMS.

Other panels highlight key features associated with EBCS. Figure 6b shows a B_z profile similar to MRX observation (double Harris sheet). The electric potential (Φ) well, strong ion outflows in the exhaust, and density increase near the separatrix are shown in Fig. 6c–e, respectively, further supporting laboratory and space observations.

Discussion

The observations of bifurcated current sheets in both laboratory (MRX) and space-based (MMS and PSP) environments suggest that expanding bifurcated current sheets (EBCS) can emerge in antiparallel magnetic reconnection in moderate β ($0.1 < \beta \lesssim 1$) plasmas regardless of system size (L): MRX with L of approximately $10d_i$, the MMS case with $L \gtrsim 10^2 d_i$, and the PSP case with $L \gtrsim 10^4 d_i$. In all three cases, the average upstream β (simple mean of values on both sides) is moderate −0.7, 0.6, and 1.2 for MRX, MMS, and PSP, respectively. EBCS are characterized by strong out-of-plane electron currents localized along the separatrix.

The apparent morphological differences between the boundaries in the solar wind and the magnetosphere can be largely attributed to the large difference in system sizes. In the heliospheric current sheet, the exhaust width is approximately $4000d_i$. Consequently, an EBCS layer of about $20d_i$ thickness appears as a sharp, step-like discontinuity on macroscopic scales, leading to its historical interpretation as an MHD slow-mode shock. In contrast, in near-Earth environments where the entire exhaust width may only be tens of d_i, the EBCS occupies a much larger relative fraction of the system, making gradients appear less abrupt. Furthermore, a recent observation shows that the localized electron diffusion region in solar wind reconnection can be bursty and turbulent, the resulting macroscopic exhaust boundaries can still propagate as coherent, long-lived structures[21].

Additionally, while the limited temporal resolution of interplanetary spacecraft has historically obscured micro-physical features like plasma waves and parallel electric fields at solar wind separatrices, high-resolution measurements have confirmed their presence. This includes observations of energetic proton beams on HCS separatrix field lines[15] and electrostatic waves localized at solar wind current sheets[52]. Therefore, rather than categorizing solar wind exhausts exclusively as macroscopic fluid shocks, our results suggest that the boundaries of prolonged reconnection across all these environments can be unified under the two-fluid kinetic framework of EBCS.

Consistent signatures across all environments include strong localized electric fields, abrupt increases in ion outflow, and density enhancements within the exhaust. The fact that these features appear in such different plasma regimes is consistent with an underlying two-fluid mechanism operating across these scales and geometries. This interpretation is further supported by large-domain VPIC simulations, which reproduce the main characteristics of EBCS and demonstrate their persistence far downstream of the X-line, consistent with both MMS and PSP observations.

An implication of these results is the role of electrostatic potential wells, arising from two-fluid effects, in driving the EBCS structure[3,25,47]. The strong in-plane electric field generates an out-of-plane electron current via the $\mathbf{E} \times \mathbf{B}$ drift, naturally forming EBCS. Our simplified momentum balance analysis shows that the potential well depth can be directly inferred from the change in reconnecting field across the separatrix, an estimate that matches MRX and MMS data. This provides a practical method to identify EBCS even when direct current measurements are unavailable, making the structure observable across both laboratory and spacecraft datasets.

Another implication of EBCS is their connection to the Petschek reconnection model. Petschek envisioned narrow, standing slow-mode shocks as the organizing structures that allow reconnection outflows to remain collimated and efficient over large distances. Although Petschek-type shocks have not been unambiguously identified in collisionless plasmas, the EBCS exhibits several characteristics expected of such shocks. It is a thin, localized structure that develops along the separatrix, is associated with an abrupt rotation of the magnetic field vector, and mediates strong ion acceleration and

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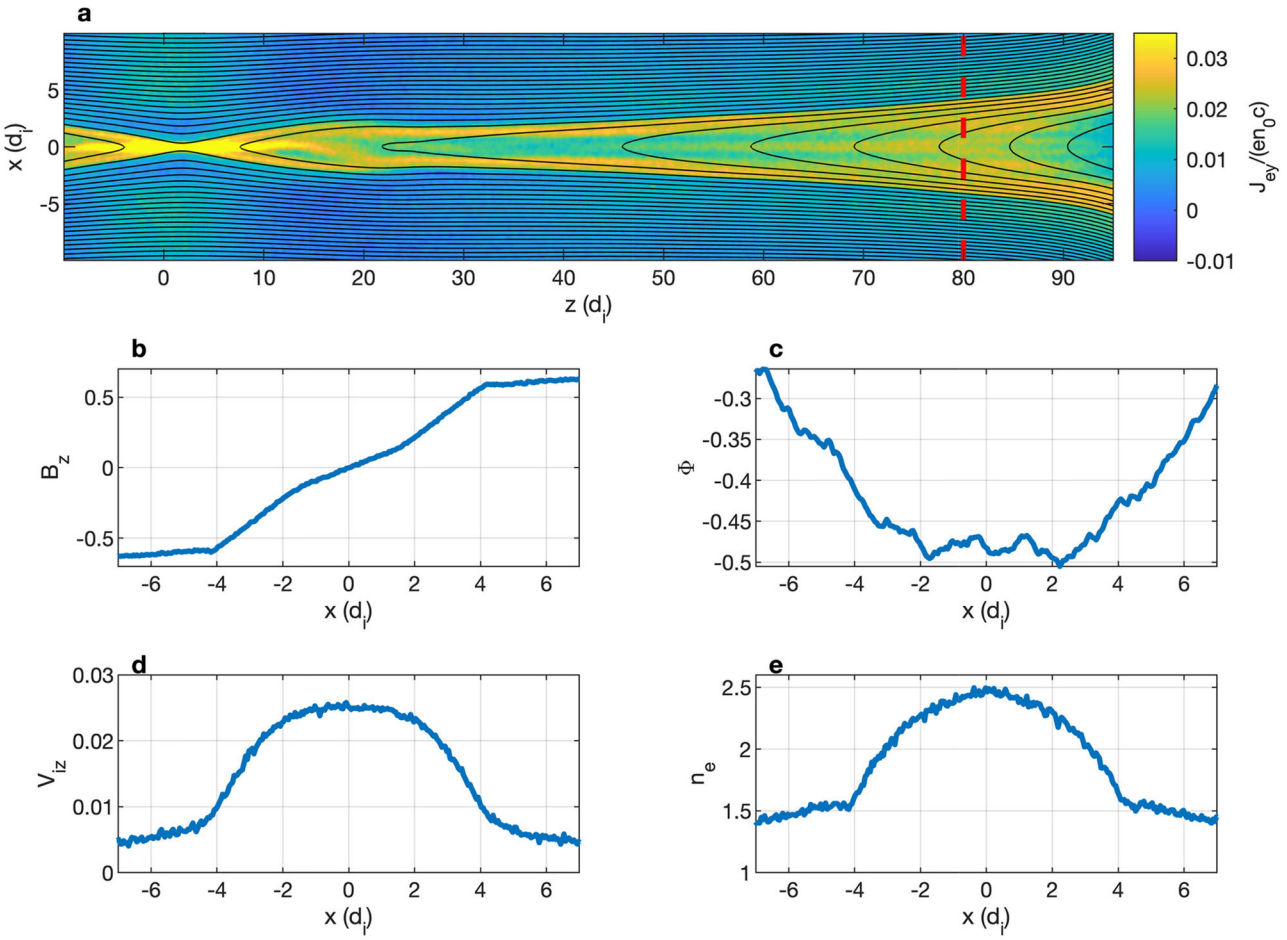


Fig. 6 | Profiles of magnetic field and plasma parameters from a 2D particle-in-cell (PIC) simulation with open boundaries along the *z* direction. All quantities are normalized, and distances are given in units of the ion inertial length d_i. **a** Out-of-plane electron current density (J_{ey}), which is similar to the total current density J_y, showing a bifurcated structure near the separatrix. The red dashed line indicates the location of the following 1D profiles. **b** Magnetic field component B_z at $z = 80d_i$, showing a profile similar to the double Harris-sheet. **c** Electric potential (Φ_0) at $z = 80d_i$. **d** Ion outflow velocity (V_{iz}) at $z = 80d_i$, showing ion acceleration across the separatrix. **e** Electron density (n_e) at $z = 80d_i$, exhibiting strong enhancements across the separatrix. All features are consistent with laboratory and space observations.

density compression. In particular, PSP data reveal that regions with clear EBCS signatures exhibit different energy partitioning, where the ion bulk flow energy exceeds the enthalpy flux. This behavior departs from the more typical dominance of thermal energy increase in MRX[40] and near-Earth space observations[39], and instead suggests that the presence of stable EBCS promotes more coherent and directed ion acceleration, making the exhaust a structure more consistent with Petschek's original framework.

The infrequent observation of EBCS in the near-Earth magnetotail can be attributed to the rapidly changing global boundary conditions of the magnetosphere, which rarely allow for prolonged quasi-steady flux annihilation. This scarcity aligns with statistical surveys characterizing bifurcated current sheets as relatively rare or atypical in the magnetotail[17]. In contrast, the global conditions of the heliospheric current sheet evolve much more slowly, allowing macroscopic boundary structures like those observed by PSP to persist, a physical reality supported by the statistically frequent observation of bifurcated sheets in the solar wind[12]. Furthermore, this steadiness depends on the plasma β. In moderate β environments (such as MRX and the PSP event), global quasi-steadiness is maintained. While exploring a comprehensive range of β in MRX is challenging due to the limited accessible parameter space (Fig. 7), making this an objective for next-generation facilities such as FLARE (Facility for Laboratory Reconnection Experiments)[53], we note that MRX discharges lacking EBCS signatures typically have unstable X-lines, supporting the importance of quasi-steady dynamics. In low-β environments typical of the magnetotail, the current sheet associated with EBCS is highly susceptible to microinstabilities like the lower hybrid drift instability (LHDI)[50]. In addition, large-scale plasmoids are formed, making the reconnection globally bursty (Fig. 8), thereby preventing the formation or persistence of coherent EBCS. When global quasi-steadiness is achieved, however, EBCS serve as robust boundary structures regardless of local bursty dynamics at the X-line.

In summary, the convergence of laboratory experiments, spacecraft measurements, and kinetic simulations shows that EBCS are a recurring signature of quasi-steady collisionless reconnection across the system sizes examined here. Their structural and energetic characteristics bear strong resemblance to the separatrix layers envisioned in the Petschek model, suggesting that EBCS may represent the kinetic-scale manifestation of Petschek-type exhaust organization. These results provide a framework for interpreting future in situ observations and designing targeted laboratory studies of reconnection energy conversion and particle acceleration.

Methods

Experimental setup

Figure 7a shows the experimental setup of the Magnetic Reconnection Experiment (MRX). The device consists of a cylindrical vacuum vessel

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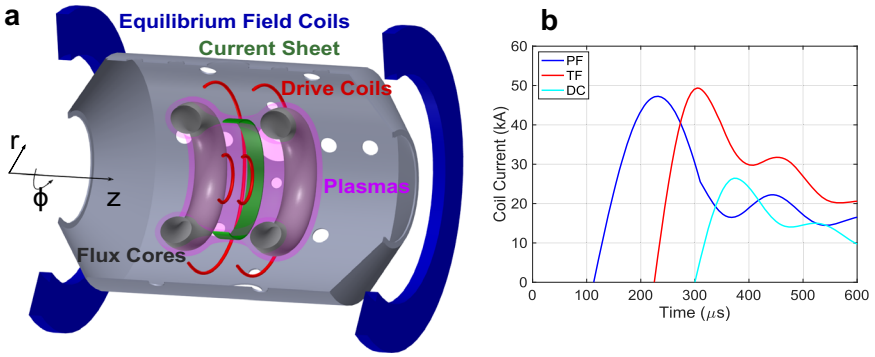


Fig. 7 | Schematic of the MRX device and coil currents. a Inside the vessel are two doughnut-shaped flux cores (gray), each containing two sets of coils: poloidal-field (PF) coils and toroidal-field (TF) coils. Drive coils (DC) provide additional reconnection drive during the quasi-steady reconnection period. The cylindrical coordinates r, ϕ, and z denote the radial, azimuthal, and axial directions, respectively. **b** Typical coil currents.

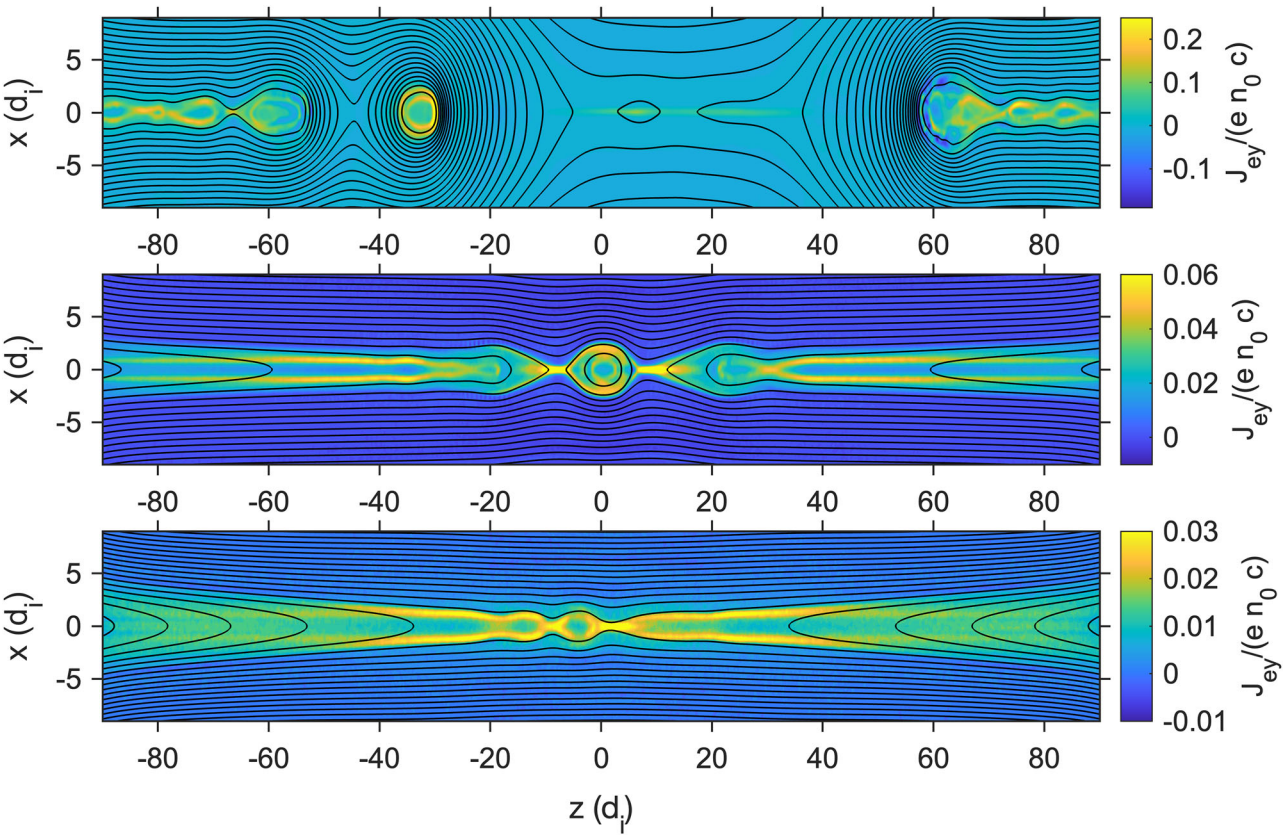


Fig. 8 | Dependence of EBCS formation and macroscopic extension on plasma β. Contour plots of the out-of-plane electron current density (J_{ey}) from a suite of strictly controlled 2D PIC simulations. All numerical and physical parameters (domain size, grid resolution, initialization, and mass/temperature ratios) are identical to the $\beta = 1.2$ simulation shown in Fig. 6, with only the total upstream plasma β varied. Spatial axes are normalized to the ion inertial length (d_i). Top: In the low-β regime ($\beta = 0.012$), the macroscopic current sheet is highly susceptible to instability. Frequent, large-scale plasmoids form and dominate the dynamics, preventing any coherent EBCS from extending downstream. Middle: At an intermediate value ($\beta = 0.12$), distinct bifurcated current sheets establish themselves and extend into the downstream exhaust region, despite frequent plasmoids still forming near the center. Bottom: In the moderate-β regime ($\beta = 0.36$), a clear EBCS forms and extends far into the downstream region (up to $z = 50d_i$).

with a diameter of 1.5 m and a length of 2 m, containing two doughnut-shaped flux cores (gray). Each flux core houses two sets of coils: poloidal-field (PF) coils and toroidal-field (TF) coils. The PF coils are wound toroidally to generate the X-line geometry at the midplane and to drive magnetic reconnection, while the TF coils are wound poloidally to inductively produce plasma around the flux cores. Additional drive coils (DC, red) are used to enhance the reconnection drive, and two external equilibrium-field (EF) coils control the position of the ring-shaped current sheet (cyan) formed at the midplane. For analysis, the cylindrical coordinate system is mapped onto a local Cartesian system, with x along the radial (r) direction and y along the azimuthal (ϕ) direction.

Figure 7b shows the coil currents for this experiment. The quasi-steady reconnection phase occurs between 320–340 μs, from which the presented data are taken, while the EF current remains constant at about 2.2 kA throughout the discharge.

Hydrogen is used as the working gas with a fill pressure of 3.5 mTorr. The PF, TF, and drive coils are operated at 11, 12, and 6 kV,

respectively, with a flux-core separation of 45 cm. To obtain one-dimensional profiles of electron density and ion outflow (Fig. 3), data from approximately 250 reproducible discharges exhibiting clear signatures of expanding bifurcated current sheets (EBCS) are averaged. Plasma conditions were verified shot-to-shot using two reference Langmuir probes placed in the upstream regions. In helium discharges under similar conditions, the upstream ion temperature is comparable to the electron temperature[31]; therefore, $T_i = T_e$ is assumed in calculating the plasma β.

The black lines in Fig. 7a represent contours of the poloidal magnetic flux, ψ, defined as

$$\psi = \int 2\pi r B_z dr, \tag{6}$$

where r is the radial coordinate and B_z is the axial magnetic-field component.

Coordinate transformation for PSP and MMS events

For the PSP event, the original measurements are reported in the *RTN* (radial, tangential, and normal) coordinate system. Following the ref. 15, we define the local *xyz* system with unit vectors z = (0.991, 0.041, 0.127)$_{RTN}$, y = (−0.043, 0.999, −0.013)$_{RTN}$, and x = (−0.126, −0.019, 0.992)$_{RTN}$. For the MMS event, a minimum variance analysis of the magnetic field in the GSE (Geocentric Solar Ecliptic) system, applied over the interval 9:46:30–9:52:00, yields z = (0.995, 0.101, −0.022)$_{GSE}$, y = (−0.031, 0.497, 0.867)$_{GSE}$, and x = (0.098, −0.862, 0.497)$_{GSE}$. The electron current density profile, $J_{ey} = -en_e v_{ey}$, shown in Fig. 4, has been low-pass filtered to suppress high-frequency noise, which is essential to resolve the peaks near the separatrix.

Additional details on Harris sheet simulations

Figures 2 and 6 show results from fully kinetic particle-in-cell (PIC) simulations using the VPIC code in two spatial dimensions with three velocity components. The initial magnetic configuration was a Harris-type current sheet with a perturbation to seed reconnection. The equilibrium field is given by

$$B_z(x,z) = B_0 \tanh\left(\frac{x}{L}\right) + \delta B_z \cos\left[\frac{2\pi(z-0.5L_z)}{L_z}\right] \sin\left(\frac{\pi x}{L_x}\right), \tag{7}$$

$$B_x(x,z) = \delta B_x \cos\left(\frac{\pi x}{L_x}\right) \sin\left[\frac{2\pi(z-0.5L_z)}{L_z}\right]. \tag{8}$$

Here, B_0 is the asymptotic magnetic-field magnitude, x and z are the normal and outflow coordinates, respectively, L is the half-width of the initial current sheet, δB_x and δB_z are the perturbation amplitudes, and L_x and L_z are the dimensions of the simulation domain in the normal and outflow directions, respectively. The mean field is therefore $B_0\hat{z}$ with a small perturbation that seeds reconnection with an X-line in the center of the domain. The plasma density profile is $n_e = n_b + n_0 \mathrm{sech}^2(x/L)$, where n_b is the background density and n_0 the peak sheet density. Both simulations used a ratio of electron plasma to cyclotron frequency of ω_{pe}/ω_{ce} = 2.0 and 2000 macroparticles per cell. For the simulation shown in Fig. 2, the domain size is $20d_i \times 20d_i$, resolved by 1280 × 1280 cells, with $m_i/m_e = 1836$, $T_{rmi}/T_e = 5$, $L/d_i = 0.5$, and $n_b/n_0 = 0.23$. For the simulations shown in Figs. 6 and 8, the domain size is $200d_i \times 40d_i$, resolved by 4032 × 800 cells, with $m_i/m_e = 100$, $T_{rmi}/T_e = 1$, $L/d_i = 5$, and $n_b/n_0 = 1.2$.

The z boundaries are treated as open[51]. At each time step, an incoming flux of macroparticles is sampled from a drifted half-Maxwellian constructed from the boundary moments (n, u, p). At the initial step, the moments are set to the equilibrium conditions. Subsequently, they are periodically updated from particles within a thin layer adjacent to the boundary to reflect the evolving outflow.

To isolate how β affects the formation and spatial extension of EBCS, we performed a strictly controlled parameter study using a new suite of VPIC simulations. In these runs, all numerical and physical parameters (domain size, grid resolution, initialization, and mass/temperature ratios) were kept exactly identical to those of the simulation shown in Fig. 6 (which successfully demonstrated macroscopic EBCS extension at $\beta = 1.2$, matching the conditions of the PSP observation). The only parameter varied across this entire suite is the upstream plasma β.

Figure 8 presents the out-of-plane electron current density for three representative cases. In the low-β regime ($\beta = 0.012$, top panel), the macroscopic current sheet is highly susceptible to plasmoid instability. Frequent, large-scale plasmoids form and dominate the dynamics, preventing any coherent EBCS from extending to the downstream region. As β increases to 0.12 (middle panel), although frequent plasmoids still form in the current sheet, distinct EBCS are now able to establish themselves and extend into the downstream exhaust region. In the moderate-β regime ($\beta = 0.36$, bottom panel), clear EBCS form and extend far into the downstream region ($z = 50d_i$).

Unlike the $\beta = 1.2$ case (Fig. 6), the EBCS in the $\beta = 0.12$ and $\beta = 0.36$ cases do not extend to the system boundaries. Before the EBCS can fully develop, plasmoids form near the center of the current sheet and propagate downstream, significantly disrupting the EBCS structure. This demonstrates that the formation and macroscopic extension of the EBCS strictly require a quasi-steady current sheet. This controlled parameter study explicitly isolates the physical conditions responsible for EBCS formation, providing evidence that moderate-to-high β plasmas promote the macroscopic extension of the EBCS by suppressing plasmoid fragmentation, thereby maintaining the quasi-steady conditions necessary for its development.

Additional details on MRX simulations

To cross-validate the experimental results, we performed fully kinetic particle-in-cell (PIC) simulations using the VPIC code in a two-dimensional cylindrical (r, z) coordinate system with three velocity components. As shown in Fig. 9, the simulations reproduce the key features observed in MRX, including the expanding bifurcated current sheets (panel a), a double-Harris-type B_z profile (panel b), a bipolar normal electric field (panel c), a sharp increase in ion outflow across the separatrix (panel d), and enhanced plasma density in the exhaust region (panel e).

The simulation geometry and boundary conditions were designed to closely reproduce the Magnetic Reconnection Experiment (MRX), which employs two flux cores. The simulation domain is a conducting box of size $L_r = 30d_i$ and $L_z = 60d_i$, (designed to encompass the entire MRX vacuum chamber), discretized into $n_r = 200$ and $n_z = 448$ cells. Note that Fig. 9 displays a zoomed-in view of the central reconnection region rather than the full computational domain. The flux cores, with outer radius 9.4 cm and inner radius 1.3 cm, were implemented as absorbing boundaries and separated by 45 cm ($18d_i$), consistent with the experimental configuration. The ion-to-electron mass ratio was set to m_i/m_e = 25, and the ratio of electron plasma frequency ω_{pe} to electron cyclotron frequency ω_{ce} was fixed at 1. Each cell contained 200 macroparticles per species. The system was advanced for 4×10^3 time steps with a normalized step size of $\Delta t\omega_{pe} = 0.24$.

The MRX poloidal field (PF) coils were modeled by prescribing, as a function of time, the out-of-plane current density within the two flux-core regions. The time dependence of the coil current was chosen to closely follow the actual PF coils in MRX[54]:

$$I_{coil}(t) = \frac{I_0}{6}\left[1 + 5\cos^2\left(\frac{\pi t}{2\tau_D}\right)\right], \tag{9}$$

where $I_{coil}(t)$ is the prescribed PF-coil current, t is time, I_0 is the current amplitude, and τ_D is the characteristic current ramp-down time. The

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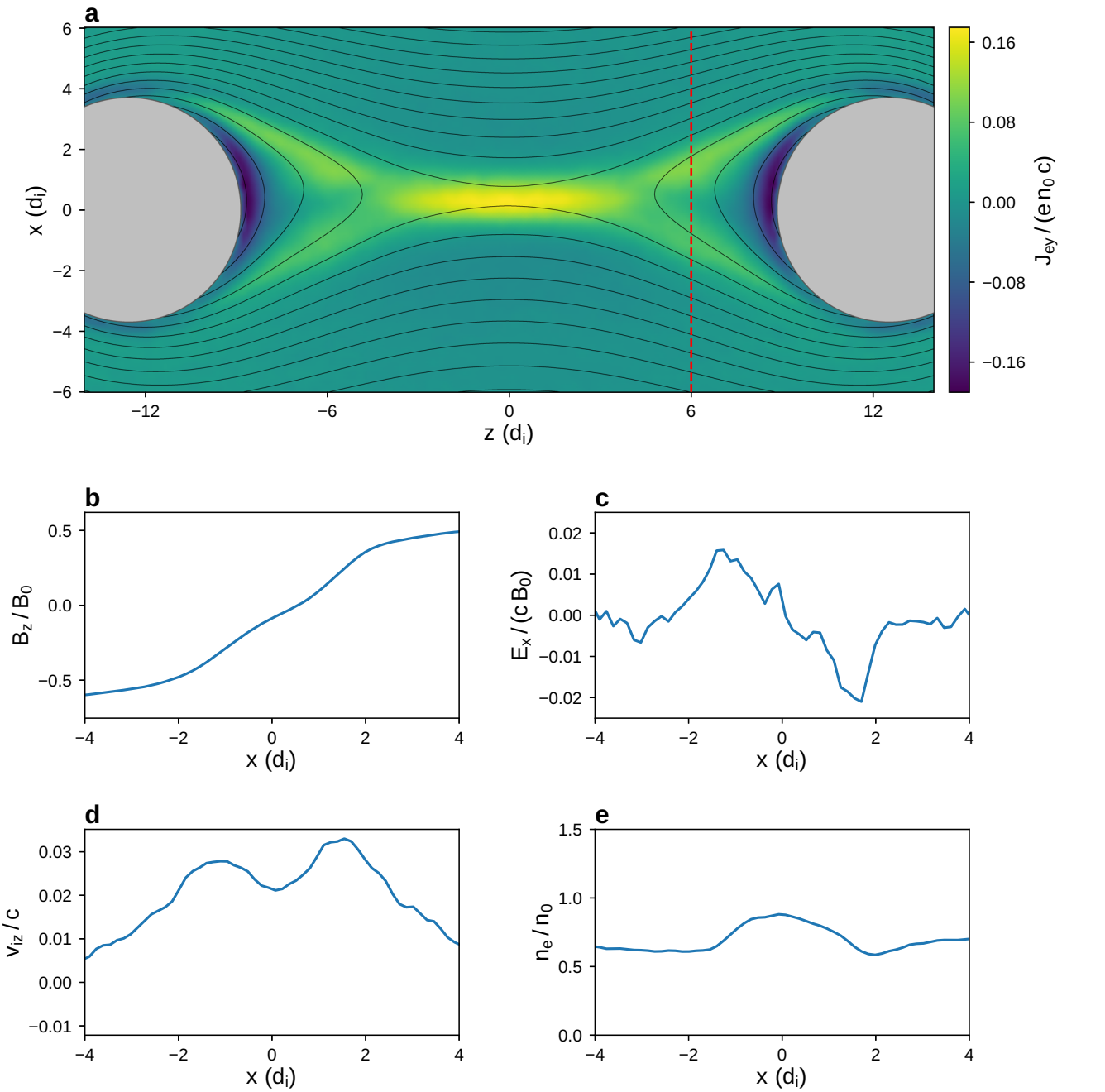


Fig. 9 | Profiles of magnetic field and plasma parameters from a 2D PIC simulation with MRX geometry. A VPIC simulation with cylindrical geometry is used to reproduce the expanding bifurcated current sheets (EBCS) observed in MRX. The vertical red dashed line in (**a**) marks $z = 6d_i$, where the 1D profiles in (**b**–**e**) are taken. All quantities are normalized. **a** Out-of-plane electron current density, with gray circles marking the location of the flux cores. **b** B_z profile at $z = 6d_i$, showing a double-Harris structure similar to that observed in the experiment. B_0 is the reference magnetic field. **c** E_x profile, displaying a bipolar structure with peaks near the separatrix, corresponding to the potential well seen in measurements. **d** Ion outflow velocity, which increases sharply across the separatrix. **e** Electron density normalized to the reference density (n_0), which peaks in the exhaust region with strong enhancements across the separatrix.

current magnitude I_0 was chosen such that the electron beta, $\beta_e = 8n_0T_0/B_0^2$, attains the desired value at a reference position located $9d_i$ upstream from the midpoint between the two flux cores at $t = 0$. The initial electron thermal velocity was determined from the specified β_e and ω_{pe}/ω_{ce}, while the ion thermal velocity was set assuming that T_e and T_i are comparable, consistent with experimental observations.

Data availability

Source data for the laboratory measurements, spacecraft analyses, and VPIC simulations underlying the figures generated during the current study have been deposited in Figshare and are publicly available at https://doi.org/10.6084/m9.figshare.30501362. The Magnetospheric Multiscale (MMS) datasets analyzed during the current study are publicly available from the MMS Science Data Center at https://lasp.colorado.edu/mms/sdc/publicunder the following data products: MMS/FGM/SRVY/L2 for survey magnetic-field data; MMS/EDP/FAST/L2/DCE for fast electric-field data; MMS/FPI/FAST/L2/DES-MOMS for

fast electron moments; and MMS/FPI/FAST/L2/DIS-MOMS for fast ion moments. The Parker Solar Probe (PSP) datasets analyzed during the current study are publicly available at https://fields.ssl.berkeley.edu/data/and https://sweap.cfa.harvard.edu/pub/data/sci/sweap/under the following data products: PSP/FIELDS/L2/mag_RTN_4_Sa_per_Cyc for magnetic-field data; and PSP/SWEAP/SPAN-I/L3/spi_sf00 for proton density, velocity, and temperature data.

Code availability

The VPIC (Vector Particle-In-Cell) code, version 1.2, used for the particle-in-cell simulations in this study is publicly available from the Los Alamos National Laboratory GitHub repository at https://github.com/lanl/vpic. Experimental and spacecraft data were analyzed using MATLAB R2024a and Python 3.12. Custom analysis codes used for data processing and figure generation are available from the corresponding author upon request.

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Author contributions

J.Y. carried out laboratory experiments, data analysis, primary interpretation of data, and manuscript writing. M.Y. and H.J. contributed to data interpretation and manuscript writing. T.R. and A.L. carried out numerical simulations and contributed to manuscript writing. T.P. contributed data interpretation. S.S. carried out numerical simulations for MRX experiments and contributed to manuscript writing. A.R. carried out numerical simulations for Fig. 8 and contributed to manuscript writing.

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Competing interests

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Correspondence and requests for materials should be addressed to Jongsoo Yoo.

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